

Human Development Index (HDI) Prediction

Project Description:

The Human Development Index (HDI) Prediction System is a Machine Learning-based web application designed to predict the Human Development Index category of a country using important socio-economic indicators. The application helps users estimate whether a country's development level falls under Low, Medium, High, or Very High Human Development based on four key input parameters: Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) Per Capita.

The system is developed using the Flask web framework and integrates a pre-trained Machine Learning model for real-time prediction. The frontend is designed using HTML and CSS, providing a simple, responsive, and user-friendly interface. Users can enter the required development indicators through the web interface, and the application instantly predicts the corresponding HDI category.

The backend loads the trained Machine Learning model and label encoder using Python's Pickle library. After validating the user inputs, the system processes the data through the trained model and displays the prediction with visually distinct color-coded results, making the output easy to understand.

Scenarios

Scenario 1: Government Development Planning

A government planning officer wants to estimate the Human Development Index category of a country after implementing improvements in healthcare and education. The officer enters the updated values for Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI Per Capita into the system. The application predicts the country's HDI category, helping policymakers evaluate the potential impact of development programs.

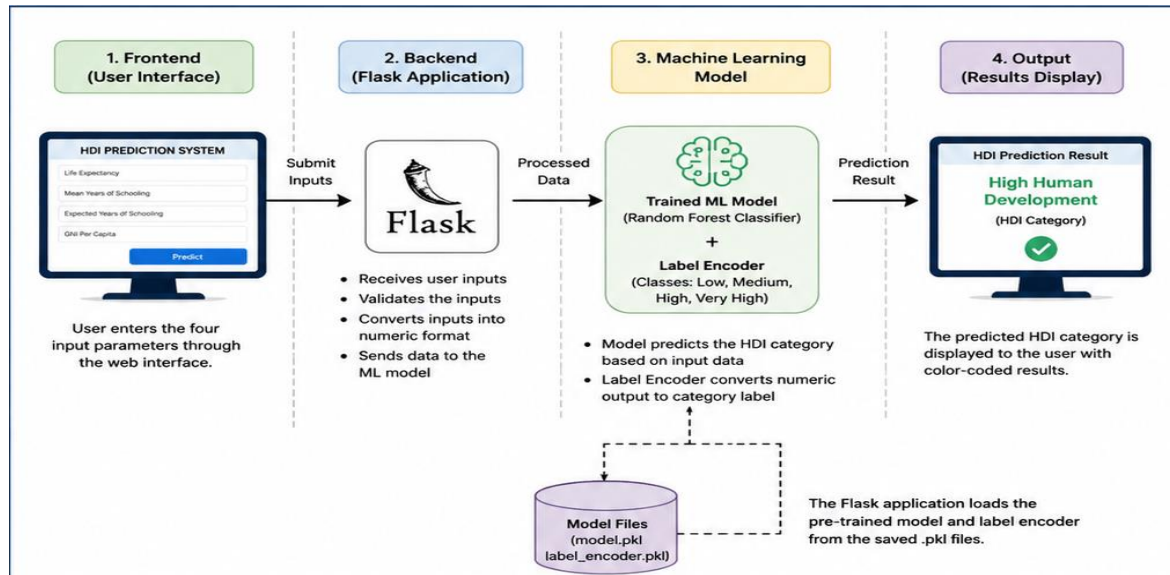
Scenario 2: Educational Research

A university student is conducting research on global human development. The student collects development indicators for multiple countries and uses the HDI Prediction System to estimate their Human Development Index categories. The predictions assist in comparing development levels among countries and understanding how socio-economic indicators influence human development.

Scenario 3: Economic and Social Analysis

An economic analyst studies the relationship between national income, education, and healthcare. By entering different combinations of development indicators into the application, the analyst observes changes in the predicted HDI category. This helps identify which factors contribute most significantly to overall human development and supports informed decision-making for future development strategies.

Technical Architecture:



Description: The Human Development Index (HDI) Prediction System follows a modular three-tier architecture. The frontend provides a user-friendly interface built with HTML and CSS. The Flask-based backend handles user input, validates the data, and communicates with the Machine Learning model. The pre-trained Random Forest model processes the input indicators and predicts the HDI category (Low, Medium, High, Very High) using the loaded model and label encoder. The prediction result is then sent back to the frontend and displayed to the user with color-coded output

Pre-requisites:

- Flask Framework Knowledge: [Flask Documentation](#)
- Python Programming Proficiency: [Python Documentation](#)
- Machine Learning Basics: [Scikit-learn Documentation](#)
- HTML, CSS, and JavaScript Skills: [W3Schools HTML/CSS Tutorials](#)
- Version Control with Git: [Git Documentation](#)
- Development Environment Setup: [Flask Installation Guide](#)
- Libraries and Tools: [NumPy Documentation](#). [Pickle Module \(Python Docs\)](#)

Project Workflow:

Milestone 1: Machine Learning Model Development

- Activity 1.1: Collect and preprocess the Human Development Index (HDI) dataset by selecting relevant development indicators such as Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per Capita.
- Activity 1.2: Analyze the dataset by handling missing values, validating data quality, and preparing the features for machine learning.
- Activity 1.3: Select an appropriate Machine Learning classification algorithm for predicting the HDI category.
- Activity 1.4: Train the Machine Learning model using the processed dataset and save the trained model and label encoder using Pickle for future predictions.

Milestone 2: Core Functionalities Development

- Activity 2.1: Develop the Flask backend to receive user inputs, validate the data, process the prediction request, and communicate with the trained Machine Learning model.
- Activity 2.2: Implement the prediction functionality that accepts user inputs, converts them into numerical format, performs prediction using the trained model, and converts the output into a readable HDI category.

Milestone 3: App.py Development

- Activity 3.1: Develop the main application logic in app.py, including Flask routing, model loading, prediction processing, exception handling, and communication between the frontend and backend.

Milestone 4: Frontend Development

- Activity 4.1: Design a responsive and user-friendly web interface using HTML and CSS for collecting user inputs and displaying prediction results.
- Activity 4.2: Create separate webpages for the project dashboard and prediction interface, ensuring smooth navigation and an attractive user experience.

Milestone 5: Deployment and Testing

- Activity 5.1: Configure the application environment by installing all required libraries and dependencies.
- Activity 5.2: Deploy the Flask application locally, verify all functionalities, test different input combinations, and validate prediction accuracy and application performance.

Milestone 6: Conclusion

- Activity 6.1: Evaluate the overall performance of the Human Development Index Prediction System, verify the correctness of prediction results, and identify possible future enhancements such as data visualization, explainable AI techniques, prediction history, and cloud deployment.

Milestone 1: Machine Learning Model Development

In this milestone, the primary objective is to develop a Machine Learning model capable of predicting the Human Development Index (HDI) category based on important socio-economic indicators. This phase includes collecting and preprocessing the dataset, selecting the appropriate machine learning algorithm, training the prediction model, and saving the trained model for deployment within the Flask web application. A well-trained model ensures accurate, reliable, and efficient prediction of HDI categories such as Low, Medium, High, and Very High Human Development.

Activity 1.1: Dataset Collection and Preprocessing

Before building the Human Development Index Prediction System, it is essential to collect a reliable dataset containing the development indicators that directly influence the HDI category. The dataset must then be cleaned and preprocessed to improve the performance and accuracy of the Machine Learning model.

Step 1 – Collect the HDI Dataset

Download the Human Development Index dataset from a trusted source such as:

- United Nations Development Programme (UNDP)
- World Bank Open Data
- Kaggle HDI Dataset
- Human Development Reports

The dataset should contain the following attributes:

- Country Name
- Life Expectancy at Birth
- Mean Years of Schooling
- Expected Years of Schooling
- Gross National Income (GNI) Per Capita
- Human Development Index Category (Target Variable)

These indicators are internationally recognized and are used by the United Nations to measure the overall development level of countries.

Step 2 – Import Required Libraries

Before loading the dataset, import the required Python libraries for data processing and Machine Learning.

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
```

These libraries are used for data manipulation, preprocessing, feature encoding, and dataset preparation.

Step 3 – Load the Dataset

Load the HDI dataset into Python using the Pandas library.

```
dataset = pd.read_csv("hdi_dataset.csv")
```

After loading the dataset, verify that it has been imported correctly by displaying the first few records.

```
dataset.head()
```

Step 4 – Explore the Dataset

Before training the Machine Learning model, analyze the dataset to understand its structure and quality.

Perform the following operations:

- Display the first five rows.
- Check the total number of records.
- Verify data types of each feature.
- Identify missing values.
- Calculate summary statistics.
- Detect duplicate records.

Example:

```
dataset.info()
```

```
dataset.describe()
```

```
dataset.isnull().sum()
```

```
dataset.duplicated().sum()
```

This analysis helps identify inconsistencies and prepares the dataset for preprocessing.

Step 5 – Data Cleaning

Data preprocessing is one of the most important steps in Machine Learning.

Perform the following preprocessing tasks:

- Remove duplicate records.
- Handle missing or null values.
- Verify numerical data consistency.
- Remove invalid or unrealistic values.
- Ensure all features are in the correct format.
- Encode the HDI category using a Label Encoder.

These preprocessing techniques improve the overall accuracy and reliability of the prediction model.

Step 6 – Select Input Features

The Human Development Index Prediction System uses the following four features as input variables:

- Life Expectancy
- Mean Years of Schooling
- Expected Years of Schooling
- Gross National Income (GNI) Per Capita

These indicators represent the three major dimensions of human development:

- Health
- Education
- Standard of Living

The selected features are provided by the user through the web application and passed to the trained Machine Learning model for prediction.

Step 7 – Split the Dataset

Finally, divide the dataset into training and testing datasets.

A commonly used split ratio is:

- 80% Training Dataset
- 20% Testing Dataset

Example:

```
X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.20,  
    random_state=42  
)
```

The training dataset is used to train the Machine Learning model, while the testing dataset evaluates its prediction accuracy on unseen data

Activity 1.2: Research and Select the Appropriate Machine Learning Model

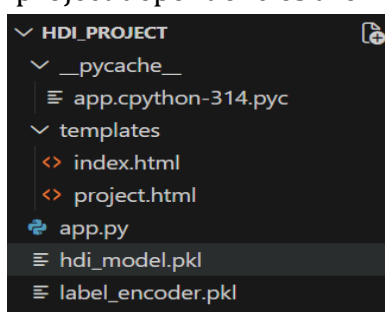
Selecting the appropriate Machine Learning algorithm is an essential step in developing the Human Development Index (HDI) Prediction System. The prediction model should accurately classify the HDI category based on important socio-economic indicators while maintaining high prediction accuracy and fast execution time. Before selecting the final model, different Machine Learning classification algorithms were researched and evaluated according to their performance, computational efficiency, and suitability for the available dataset.

Activity 1.3: Define the Architecture of the Application

The Human Development Index (HDI) Prediction System follows a simple three-tier architecture consisting of the **Presentation Layer (Frontend)**, **Application Layer (Backend)**, and **Machine Learning Layer**. This architecture ensures smooth communication between the user interface, Flask application, and the trained Machine Learning model. The user enters the required development indicators through the web interface, the backend processes the request, and the trained model predicts the corresponding HDI category.

Activity 1.4: Set Up the Development Environment

Before developing and running the Human Development Index (HDI) Prediction System, it is important to configure the development environment correctly. This includes installing Python, creating a virtual environment, installing the required libraries, and organizing the project files. A properly configured environment ensures that the application runs smoothly and all project dependencies are managed efficiently.



Milestone 2: Core Functionalities Development

The second milestone focuses on implementing the core functionalities of the Human Development Index (HDI) Prediction System. In this phase, the Machine Learning model is integrated with the Flask application, allowing users to submit development indicators and receive real-time HDI predictions. The backend processes user inputs, performs prediction using the trained model, and returns the predicted HDI category through the web interface.

Activity 2.1: Develop the Core Prediction Functionality

The core functionality of the HDI Prediction System is to predict the Human Development Index category based on four important socio-economic indicators. The application accepts user input, validates the data, converts it into a format suitable for the Machine Learning model, and displays the prediction instantly.

Milestone 3: App.py Development

Milestone 3 focuses on developing the app.py file, which serves as the core of the Human Development Index (HDI) Prediction System. This file manages the communication between the user interface and the Machine Learning model. It is responsible for handling user requests, validating input values, loading the trained model, generating predictions, and displaying the results on the web page.

Activity 3.1: Writing the Main Application Logic in app.py

The app.py file is the main entry point of the application. It initializes the Flask framework, loads the trained Machine Learning model and Label Encoder, defines the application routes, and processes prediction requests.

Define the Flask Application

The Flask application is initialized, and the required libraries are imported. The trained Machine Learning model and Label Encoder are loaded using the Pickle library.

```
from flask import Flask, render_template, request
import pickle
import numpy as np

app = Flask(__name__)

model = pickle.load(open("hdi_model.pkl", "rb"))
label_encoder = pickle.load(open("label_encoder.pkl", "rb"))
```

Define the Application Routes

The application contains three main routes that perform different functions.

Home Route

The home route displays the project dashboard.

```
@app.route("/")
def home():
    return render_template("index.html")
```

Project Route

This route opens the prediction interface where users enter the development indicators.

```
@app.route("/launch")
def launch():
    return render_template("project.html")
```

Prediction Route

The prediction route receives the user input, validates the values, processes the data, and generates the HDI prediction.

```
@app.route("/predict", methods=["GET", "POST"])
def predict():
```

Processing User Input

When the user submits the prediction form, the Flask application performs the following operations:

- Reads the values entered by the user.
- Validates that all fields contain valid numerical values.
- Converts the values into floating-point numbers.
- Creates a NumPy array containing the input features.

Example:

```
features = np.array([[
    life_expectancy,
    mean_years_schooling,
    expected_years_schooling,
    gni_per_capita
]])
```

Machine Learning Prediction

After preparing the feature array, the trained Machine Learning model predicts the Human Development Index category.

The predicted numerical value is converted into the corresponding category using the Label Encoder.

```
prediction = model.predict(features)
result = label_encoder.inverse_transform(prediction)[0]
```

The application then sends the prediction result to the HTML page for display.

Error Handling

To improve reliability, the prediction process is enclosed within a **try-except** block. If invalid data or unexpected errors occur, the application displays an informative error message instead of terminating unexpectedly.

This improves the stability and usability of the HDI Prediction System.

Milestone 4: Frontend Development

Milestone 4 focuses on designing and developing a user-friendly, responsive, and visually appealing interface for the Human Development Index (HDI) Prediction System. The frontend is developed using HTML5 and CSS3, providing an intuitive environment where users can enter development indicators and instantly view the predicted HDI category. The interface is designed to improve usability while maintaining a clean and professional appearance across different devices.

Activity 4.1: Designing and Developing the User Interface

The frontend of the application serves as the interaction layer between the user and the Machine Learning prediction system. It provides a simple navigation flow, allowing users to access the prediction tool and obtain results efficiently.

Design the Homepage

The homepage acts as the landing page of the application and provides an overview of the Human Development Index Prediction System. It introduces the project, explains its purpose, and includes a button that redirects users to the prediction page.

The homepage contains:

- Project Title
- Project Description
- Project Goal
- Importance of HDI Prediction
- "Open Predictor" Button

This layout helps users understand the purpose of the application before using the prediction system.

Activity 4.2: Creating Dynamic Content Rendering

The Human Development Index (HDI) Prediction System dynamically displays prediction results based on the values entered by the user. The Flask backend processes the submitted data, performs prediction using the trained Machine Learning model, and returns the result to the frontend. The prediction page updates automatically and displays the predicted HDI category without requiring users to manually navigate between different pages.

Milestone 5: Deployment

Milestone 5 focuses on preparing the Human Development Index (HDI) Prediction System for execution in a local development environment. This stage includes configuring the Python environment, installing the required libraries, verifying the project structure, and running the Flask application. Proper deployment ensures that the Machine Learning model, backend, and frontend work together to provide accurate real-time predictions.

Activity 5.1: Preparing the Application for Local Deployment

Before running the application, all required software, libraries, and project files must be properly configured.

Set Up a Virtual Environment

- A virtual environment is used to isolate project dependencies from other Python projects installed on the computer.
- Open Command Prompt or Visual Studio Code Terminal, then create a virtual environment using the following command:

```
python -m venv myenv
```

Activate the virtual environment.

Activity 5.2: Local Testing and Verification

After configuring the development environment and installing all required dependencies, the next step is to test the Human Development Index (HDI) Prediction System locally. Local testing ensures that the Flask application, Machine Learning model, and user interface work correctly before deployment.

Start the Flask Development Server

Activate the virtual environment and run the Flask application using the following command:

```
python app.py
```

If the application starts successfully, the terminal displays a message similar to:

```
* Running on http://127.0.0.1:5000/  
* Debug mode: On
```

This indicates that the Flask server is running successfully and is ready to receive user requests.

Open the Application in a Web Browser

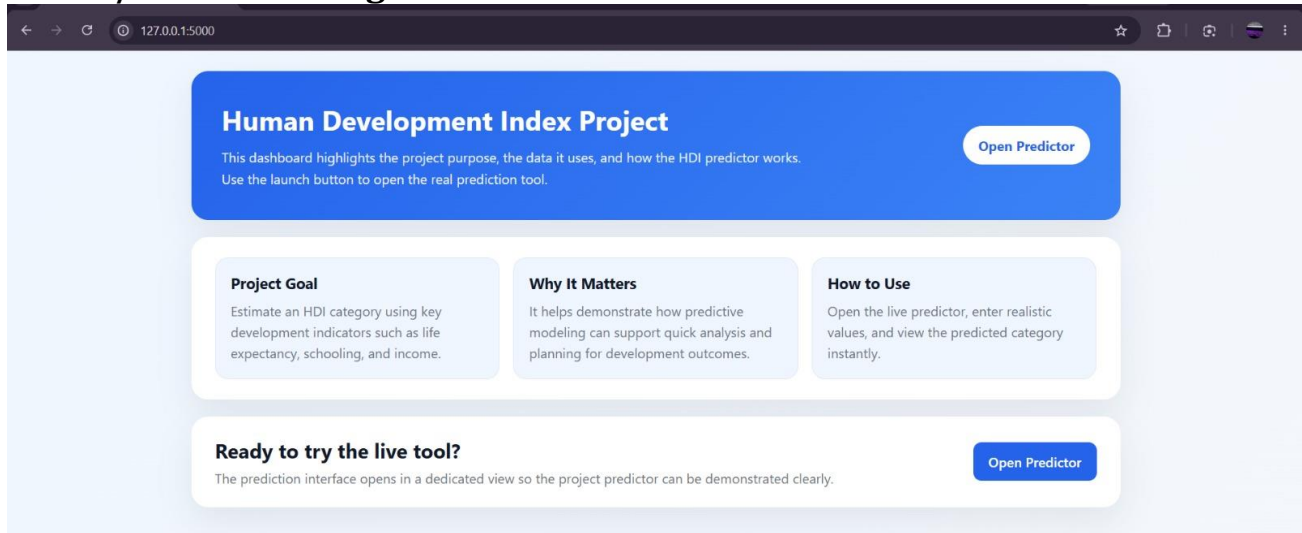
Launch a web browser such as **Google Chrome**, **Microsoft Edge**, or **Mozilla Firefox**, and open the following address:

```
http://127.0.0.1:5000/
```

The homepage of the **Human Development Index (HDI) Prediction System** will appear. From there, click the **Open Predictor** button to access the prediction interface.

Exploring the Website's Web Pages:

Home / Generator Page:

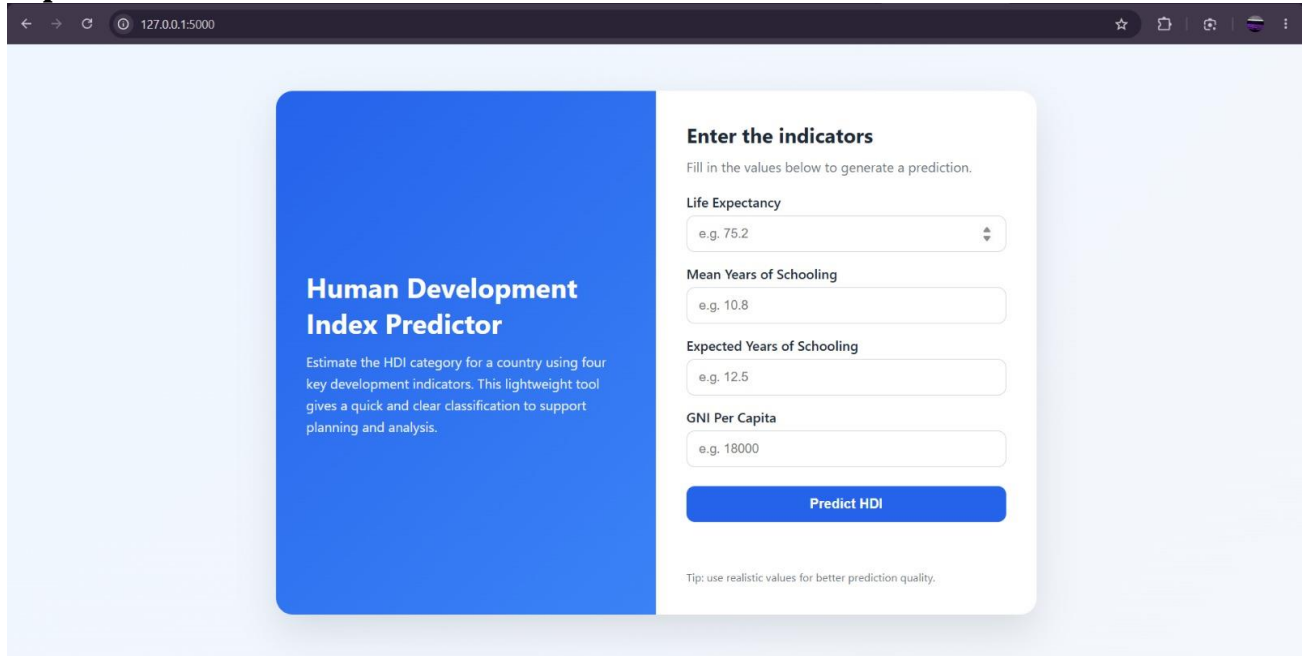


Description: The **Home Page** serves as the landing page of the Human Development Index (HDI) Prediction System. It introduces users to the purpose of the application and provides an overview of the project. The page is designed with a modern, responsive interface using HTML and CSS, making it easy to navigate and visually appealing.

The homepage contains a brief introduction explaining how Machine Learning is used to predict the Human Development Index category based on socio-economic indicators. It also includes a navigation button that redirects users to the prediction interface.

The design focuses on simplicity and usability, ensuring that users can quickly understand the project before accessing the prediction module.

Input Section:

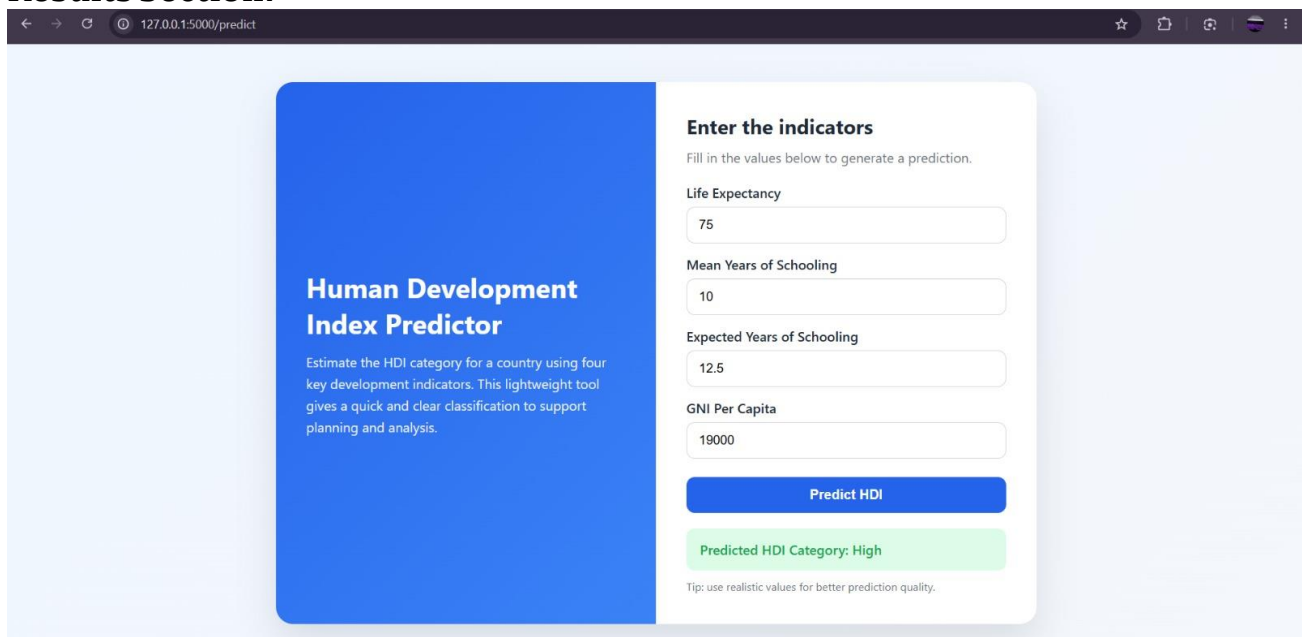


The screenshot shows the input section of the Human Development Index Predictor. It features a blue sidebar with the title "Human Development Index Predictor" and a description: "Estimate the HDI category for a country using four key development indicators. This lightweight tool gives a quick and clear classification to support planning and analysis." The main content area is white and contains a form titled "Enter the indicators" with the instruction "Fill in the values below to generate a prediction." The form has four input fields: "Life Expectancy" (with a placeholder "e.g. 75.2"), "Mean Years of Schooling" (with a placeholder "e.g. 10.8"), "Expected Years of Schooling" (with a placeholder "e.g. 12.5"), and "GNI Per Capita" (with a placeholder "e.g. 18000"). Below the fields is a blue "Predict HDI" button. At the bottom, a tip reads: "Tip: use realistic values for better prediction quality."

Description: The **Prediction Page** is the primary interface of the Human Development Index (HDI) Prediction System, where users interact with the Machine Learning model. This page allows users to enter the required socio-economic indicators and receive an instant prediction of the Human Development Index category. The interface is designed to be responsive, user-friendly, and visually appealing, ensuring a smooth experience across desktop and mobile devices.

The prediction form collects four important development indicators that directly influence the Human Development Index. Once the user submits the form, the Flask backend processes the data, performs prediction using the trained Machine Learning model, and displays the predicted HDI category on the same page.

Results Section:



The screenshot shows the results section of the Human Development Index Predictor. It features the same blue sidebar as the input section. The main content area is white and contains the same "Enter the indicators" form, but with the values entered in the input fields: "Life Expectancy" (75), "Mean Years of Schooling" (10), "Expected Years of Schooling" (12.5), and "GNI Per Capita" (19000). Below the "Predict HDI" button, a green box displays the result: "Predicted HDI Category: High". At the bottom, the same tip is present: "Tip: use realistic values for better prediction quality."

Description: After the prediction process is completed, the application displays the output directly below the input form. The result section includes:

- Predicted Human Development Index (HDI) Category.
- Color-coded result message for easy interpretation.
- Previously entered input values.
- A clear indication of whether the predicted category is **Low, Medium, High, or Very High Human Development**.

This approach allows users to review their submitted values and perform additional predictions without re-entering all the information.

Conclusion:

The Human Development Index (HDI) Prediction System is a Machine Learning-based web application developed to predict the Human Development Index category of a country using important socio-economic indicators such as Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) Per Capita. The system successfully integrates Machine Learning with the Flask web framework to provide real-time predictions through a simple, responsive, and user-friendly interface.

The application demonstrates how Artificial Intelligence and Machine Learning can support the analysis of human development by transforming statistical indicators into meaningful predictions. The trained Machine Learning model processes user-provided data efficiently and classifies the country into one of the Human Development Index categories: Low, Medium, High, or Very High Human Development. The prediction results are displayed instantly with color-coded indicators, improving readability and user experience.

Throughout the project, all major development phases—including dataset collection, data preprocessing, Machine Learning model development, Flask backend implementation, frontend interface design, testing, and deployment preparation—were completed successfully. The integration of HTML, CSS, Python, Flask, NumPy, and Scikit-learn demonstrates the practical application of modern web technologies with Machine Learning techniques.

The Human Development Index Prediction System can serve as a valuable educational and analytical tool for students, researchers, policy analysts, and government organizations interested in understanding the relationship between education, healthcare, income, and overall human development. It provides a simple method for estimating the development category of a country based on key indicators and can support research, planning, and awareness activities.

Future improvements can further enhance the application by incorporating interactive dashboards, graphical visualization of prediction results, prediction history using a database, user authentication, PDF report generation, cloud deployment, and Explainable Artificial Intelligence (XAI) techniques such as SHAP or LIME to improve transparency and interpretability of the Machine Learning model. Integration with real-time datasets from organizations such as the United Nations Development Programme (UNDP) or the World Bank would also improve prediction accuracy and practical usability.

In conclusion, the Human Development Index Prediction System successfully demonstrates the effective integration of Machine Learning and web technologies to solve a real-world problem. The project highlights the importance of predictive analytics in evaluating human development and provides a scalable foundation for future enhancements and advanced analytical applications.